

Analog IC Design Implementation and Verification

Week 2 – Amplifiers

Version IC 23.1 ISR11.29 with SPECTRE 24.1 ISR1

Revision 1.0

Estimated time: 1 Week

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Module Objectives

In this module, you will learn how to design and analyze CMOS structures, such as amplifiers, mirrors, multistage amplifiers, comparators, tuned amplifiers, and differential amplifiers.

In this module, you will learn how to

- Describe four kinds of configurations for small- and large-signal amplifiers
- Describe various kinds of active loads and current sinks in amplifiers
- Describe current mirrors and compare different kinds
- Describe the design considerations for differential amplifiers
- Compare the design and function of cascode, cascade, and folded cascode amplifiers
- Compare and be able to provide examples of the four feedback topologies



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Module Objectives (continued)

You will also learn how to

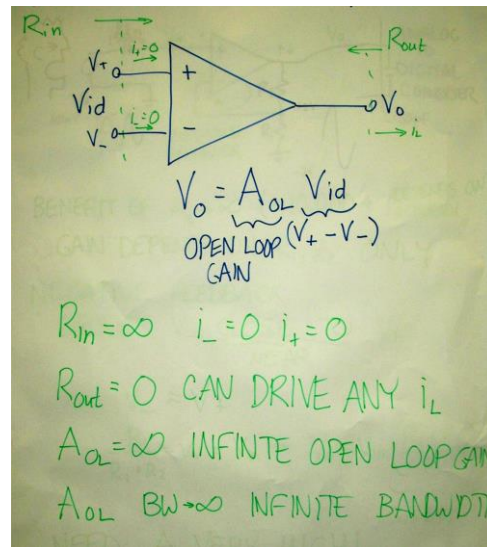
- Describe the structure and performance of various kinds of comparators
- Describe different kinds of compensation in amplifiers
- Explain the importance of settling time in the design of circuits and describe the factors that affect settling time in circuits <objective 10>
- Describe the main classes of amplifiers
- Compare the advantages and disadvantages of the main methods for tuning amplifiers
- Explain the challenges of and provide examples for setting the output common-mode voltage



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Ideal Opamp Properties

- Analog IC design goal: Design circuit to get as close to ideal as necessary
- Determined by the requirements of the application

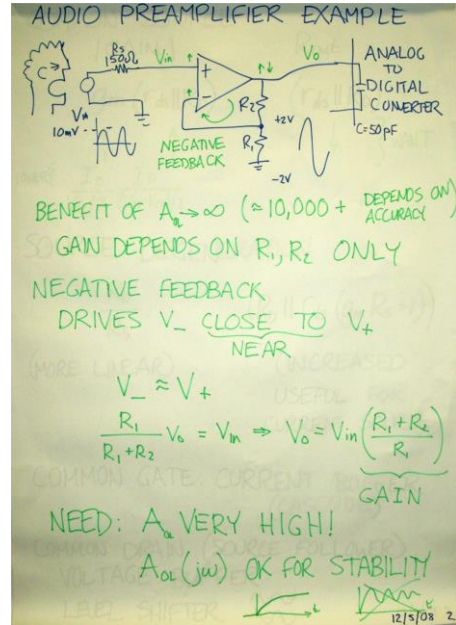


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Example Application: Audio Preamplifier

- Microphone signal source:
10 mV amplitude, $R_S = 150\Omega$
- A/D converter ILoad:
2V range, $C_L = 50\text{ pF}$
- Need amplifier to increase signal amplitude, drive C_L
- Requirements on opamp open loop gain A_{OL}
- Large A_{OL} : Closed-loop amplifier gain linear, determined by R_1, R_2

Caution: Conditions on the transfer function of $AOL(j\omega)$ for closed-loop stability.



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Single MOS Amplifiers

Small-signal amplifier configurations

- Common source
- Common gate
- Common drain
- Common source with source degeneration

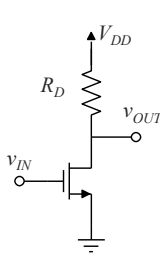


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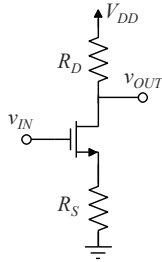
Amplifier Configurations

Different configurations are needed for different purposes:

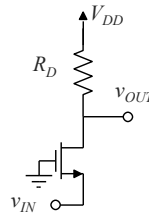
- Common source (with and without source degeneration): high voltage gain, high input resistance
- Common gate: unity current gain, low input resistance
- Common drain: near unity voltage gain, low output resistance



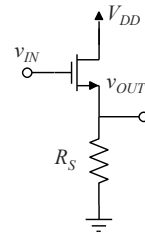
Common source



Source degeneration



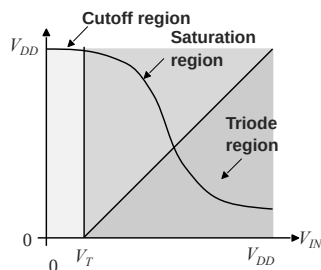
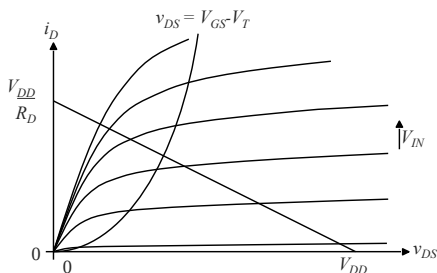
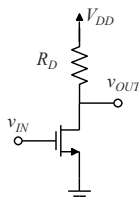
Common gate



Common drain

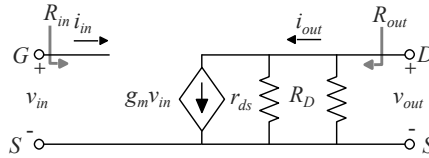
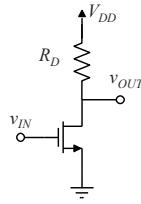
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Common-Source Large-Signal Models



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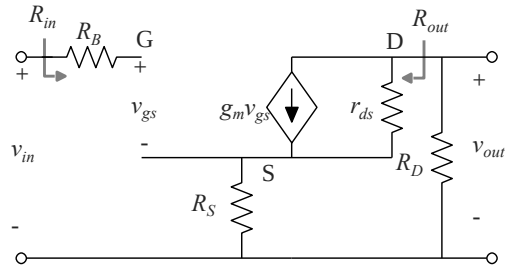
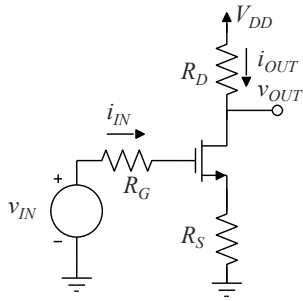
Common-Source Small-Signal Models



$$R_{in} = \infty \quad R_{out} = \frac{r_{ds} R_D}{r_{ds} + R_D} \quad \frac{v_{out}}{v_{in}} = \frac{-g_m \cdot r_{ds} \cdot R_D}{r_{ds} + R_D} \quad \frac{i_{out}}{i_{in}} = \infty$$

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Common-Source with Source Degeneration



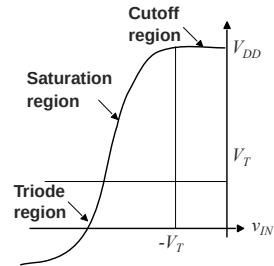
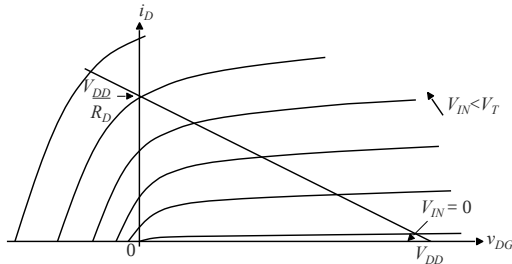
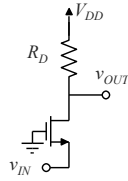
$$R_{in} = \infty$$

$$R_{out} = r_{ds} [1 + (g_m + g_{mbs}) R_S] + R_S \approx r_{ds} g_m R_S$$

$$\frac{v_{out}}{v_{in}} \approx -\frac{g_m R_D}{1 + g_m R_S} \rightarrow -\frac{R_D}{R_S} \Big|_{\text{maximum gain}}$$

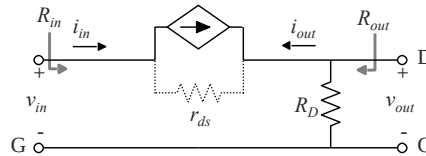
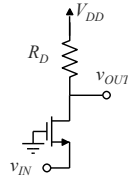
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Common-Gate Large-Signal Models



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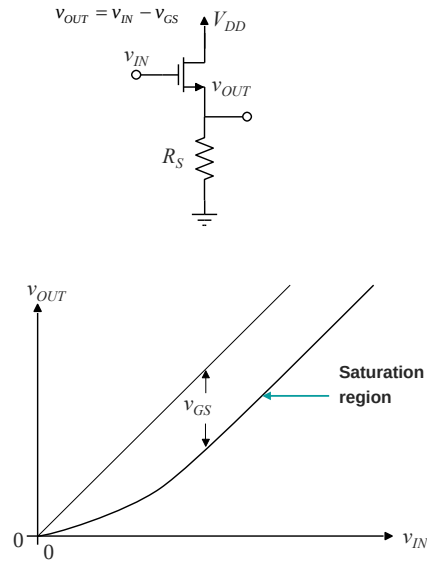
Common-Gate Small-Signal Models



Neglecting r_{ds} : $R_{in} \approx \frac{1}{g_m}$ $R_{out} \approx R_D$ $\frac{v_{out}}{v_{in}} \approx g_m R_D$ $\frac{i_{out}}{i_{in}} = -1$

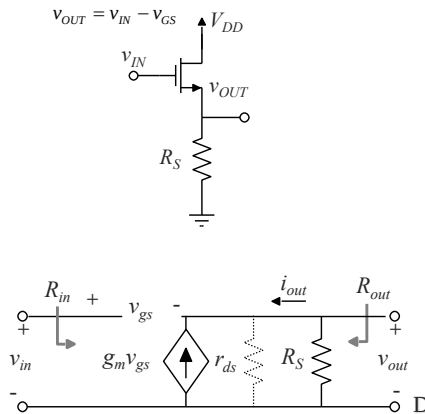
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Common-Drain Large-Signal Models



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Common-Drain Small-Signal Models



$$\text{Neglecting } r_{ds}: R_{in} = \infty \quad R_{out} = R_S \left\| \left(\frac{1}{g_m} \right) = \frac{R_S}{1 + g_m R_S} \quad \frac{v_{out}}{v_{in}} = \frac{g_m R_S}{1 + g_m R_S} < 1$$

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Summary: MOS Single-Stage Amplifiers

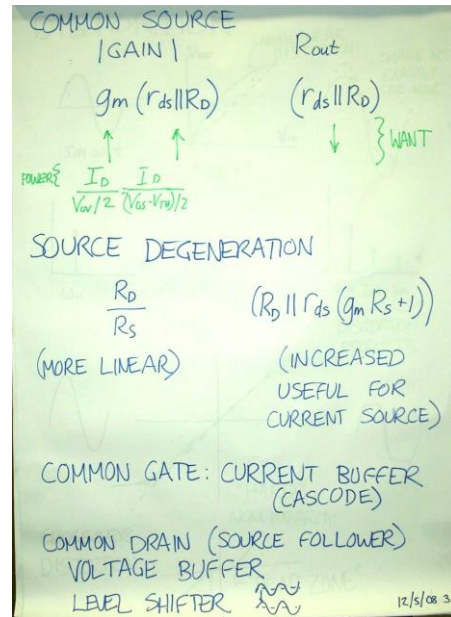
Small-Signal Performance	Common Source	Common Gate	Common Drain
Input resistance	∞	$\frac{r_{ds} + R_D}{1 + g_m r_{ds}}$	∞
Output resistance	$\frac{r_{ds} R_D}{r_{ds} + R_D}$	$\frac{r_{ds} R_D}{r_{ds} + R_D}$	$\frac{R_S}{1 + g_m R_S}$
Voltage gain	$\frac{-g_m r_{ds} R_D}{r_{ds} + R_D}$	$g_m R_D$	0.8
Current gain	∞	-1	∞



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Observations on One-MOSFET Amplifiers

- Common source tradeoff is reducing R_{out} (good for voltage amplifier) to decrease gain (bad for opamp).
- Source degeneration reduces gain and makes gain more linear. Increases R_{out} , which is useful in current source design.
- Common gate amplifier acts as a current buffer (a useful concept in cascode).
- Common drain (also called *source follower*) can be understood as either of the following:
 - Voltage buffer
 - Level shifter (useful between amplifier stages that have different bias points)



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Introduction: Amplifier Classes

- Efficiency of amplifiers is often very important:
 - Battery-operated devices
 - High-powered devices that need to be cooled
- Simple calculations for sine-wave signals are often optimistic compared to real-world signals.
- The price of higher efficiency is complexity and distortion.



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Definition of Distortion in Amplifiers

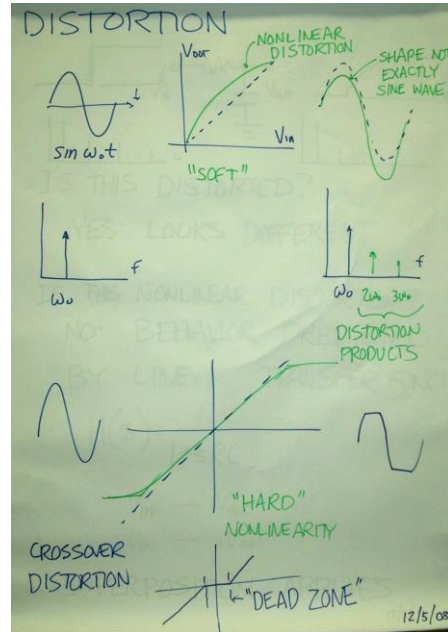
Nonlinear distortion: nonlinearity in input-output characteristic.

Examples

- **Soft distortion:** smooth curve.
- **Hard distortion:** clipping at limits.
- **Crossover distortion:** dead zone near zero.

All create harmonics (distortion products) in the frequency domain.

Key point: Linear system can produce only those output frequencies that were in the original input.



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Terminology: Linear Distortion

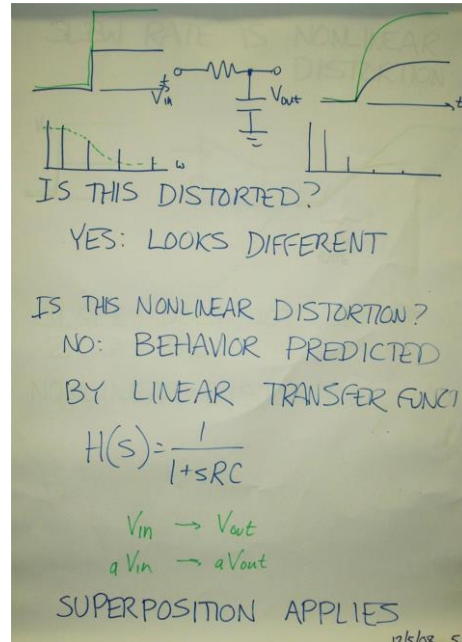
- Consider a step input applied to an RC lowpass filter.
- Output is an exponential.

Is this distorted? Yes.

- Looks different; waveform properties, such as rise time, are different.

Is this nonlinear distortion? No.

- Behavior is accurately described by linear system theory. No new frequency components in the output.
- Superposition applies: Different components of input waveform in the frequency domain are scaled and time-shifted by the transfer function.
- If the input step is scaled, the output waveform shape is also scaled by the same factor.

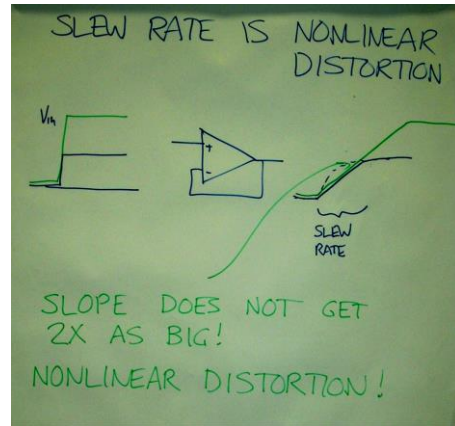


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Slew Rate Limiting as Nonlinear Distortion

- Slew rate is an example of nonlinear distortion.
- If the input step size is scaled up, output dV/dt does not increase.
- Example of nonlinear behavior.

Important to know from observed (or simulated) behavior whether the problem is caused by *linear* or *nonlinear* effects. This is a clue as to what design techniques are available to fix the problem.



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Class A

- If the quiescent current is higher to reduce distortion, the efficiency is lower.
- If the output signal is smaller than the absolute maximum, the efficiency is lower.
- If the V_{DD} is not much larger than V_{dsat} then the efficiency is lower.
- For real applications the 25% maximum is quite unrealistic.
- Class A is defined as a mode of operation where the output stage draws a constant supply current independent of load current and the devices never shut off.
- Class A is the lowest distortion and simplest of the classes.



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Class A (continued)

$$V_{out_{RMS}} = \frac{V_{DD} - 2V_{D(sat)}}{2\sqrt{2}}, \quad P_{out} = \frac{V_{RMS}^2}{R_L}$$

$$\therefore P_{out} = \frac{(V_{DD} - 2V_{D(sat)})^2}{8R_L}$$

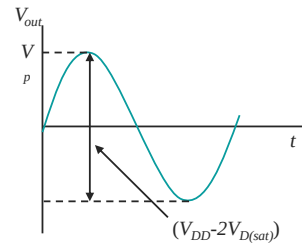
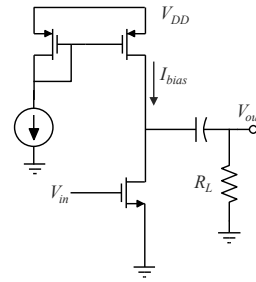
$$I_{bias} \geq \frac{V_p}{R} = \frac{V_{DD} - 2V_{D(sat)}}{2R}$$

$$P_{in} \geq V_{DD} I_{bias} = \frac{V_{DD}(V_{DD} - 2V_{D(sat)})}{2R_L}$$

$$\varepsilon = \frac{P_{out}}{P_{in}} = \frac{2R_L(V_{DD} - 2V_{D(sat)})^2}{8R_L V_{DD}(V_{DD} - 2V_{D(sat)})} = \frac{V_{DD} - 2V_{D(sat)}}{4V_{DD}}$$

For minimum I_{bias} & $V_{DD} \gg 2V_{D(sat)}$:

$$\varepsilon = \frac{1}{4}$$



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Class B

- Class B is defined as conduction for 50% of the cycle.
- The large distortion that would result from deleting half of the signal is avoided by having two half circuits share the work, each operating at a higher efficiency. This is known as the *push-pull operation*.
- In some RF applications, the second half of the waveform can be supplied from a resonant tank, so push-pull is unnecessary.



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Class B (continued)

- Class B operation also brings with it the problem of crossover distortion. If M1 and M2 do not hand off to each other perfectly, the output waveform is distorted. This can be from a dead time (M1 and M2 off), too much or too little gain in the crossover region from M1 and M2 behaving differently at very low currents, or from time delays where it takes one of the transistors some time to go from cutoff to on.
- One solution is to leave both transistors on at a low level (Class AB) so the handoff can be cleaner.
- Accurately controlling the bias level can be difficult over process and temperature because the output transistors are often sized very differently from the bias network and self-heating from the high output current levels.



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Class B (continued)

$$P_{out} = \frac{(V_{DD} - V_{D(sat)})^2}{2R_L}$$

Power from V_{DD}^+ is same as power from V_{DD}^- .

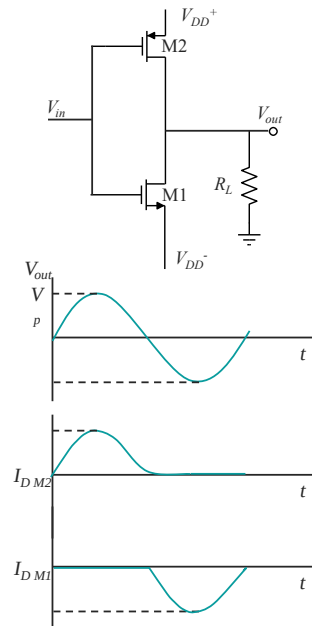
From V_{DD}^+ :

$$P_{in} = V_{DD} \frac{1}{T} \int_{t=0}^{t=T} I_{DM2} dt$$

$$P_{in} = V_{DD} \frac{1}{T} \int_{t=0}^{t=T/2} \frac{V_p}{R} \sin\left(\frac{2\pi t}{T}\right) dt$$

$$P_{in} = V_{DD} \frac{V_p}{2\pi R} \left(-\cos\left(\frac{\pi t}{T}\right) \right) \Bigg|_0^{T/2} = \frac{V_p V_{DD}}{\pi R}$$

$$P_{in} = \frac{(V_{DD} - V_{D(sat)}) V_{DD}}{\pi R}$$



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Class B (continued)

$$P_{inTotal} = \frac{2V_{DD}(V_{DD} - V_{D(sat)})}{\pi R_L}$$

$$\frac{P_{out}}{P_{in}} = \frac{\pi R_L (V_{DD} - V_{D(sat)})^2}{2R_L 2V_{DD} (V_{DD} - V_{D(sat)})}$$

$$\varepsilon = \frac{P_{out}}{P_{in}} = \frac{\pi (V_{DD} - V_{D(sat)})}{4V_{DD}}$$

For $V_{DD} \gg V_{D(sat)}$

$$\varepsilon \approx \frac{\pi}{4} \text{ or } 78.5\%$$

Again, this is only for the ideal case of a large input signal and high V_{DD} .

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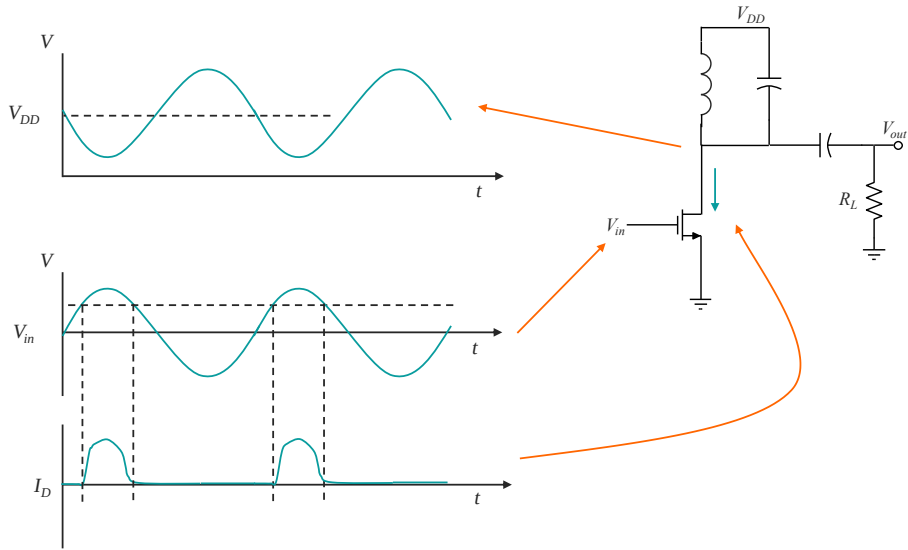
Class C

- Class C operation is when the output stage is on less than 180 degrees of the cycle.
- There is no way to have multiple stages split up the work to create the desired output waveform, so this is limited to RF amplifiers which can use a resonant tank to restore the waveform.
- The efficiency of a class C amplifier is even higher than a class B amplifier because it avoids some of the low efficiency time when the drive transistor is only partially on.
- The linearity of a class C amplifier is very poor. For small input signals, the gain drops to zero. This class is most useful for the generation of fixed amplitude (CW) signals that have limited fractional bandwidth and little or no amplitude variation.



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Class C (continued)



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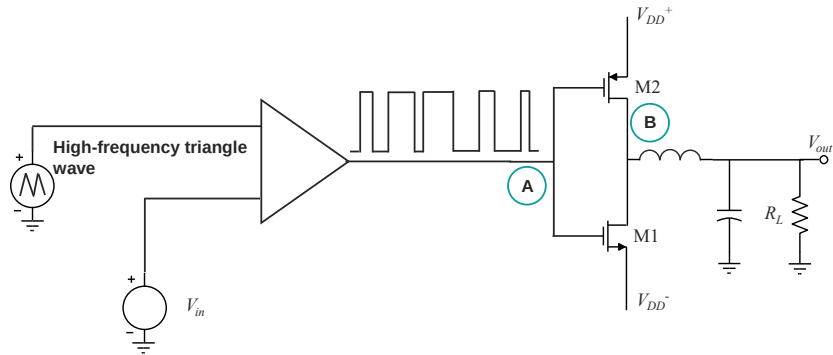
Class D

- Input signal is compared to a high-frequency triangle wave.
- The comparator output is a pulse train at the triangle-wave frequency with a duty cycle proportional to the amplitude of the input signal.
- The average value of the pulse train is proportional to the input signal both at the comparator output and the higher power version after the output stage.
- The output stage transistors operate as switches and dissipate very little power because they have either a large current and a small voltage across them or no current and a large voltage across them.
- The output-filter losses are only the ESR of the inductor and the capacitor. The low-pass filter removes the high-frequency content and only passes the *average value* or low-frequency power to the load.
- This is an idealized design to explain class D. Practical class D amplifiers are more complex to lower distortion and filter losses.



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Class D (continued)



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Other Classes

- Other classes are rarely used but can be useful for specialized applications.
- Class E and F are specialized RF amplifier techniques to improve efficiency over class C. They reduce harmonic losses in the output network to improve efficiency.
- Class G is similar to class AB but uses multiple power supplies to achieve better efficiency with less than full-scale signals.



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Introduction: Active Loads

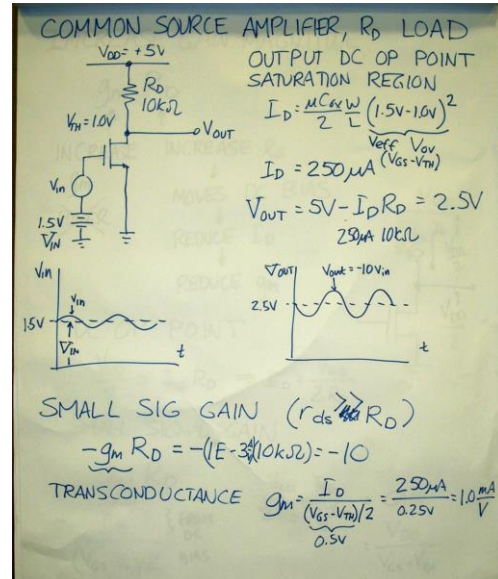
- Active loads
 - Large-signal behavior
 - Small-signal behavior
 - Frequency response
- Current sinks
 - Simple
 - Cascode
 - High-swing cascode
 - Improved cascode
 - Self-biased cascode
 - Regulated cascode



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Increasing Gain of Common Source Amplifier

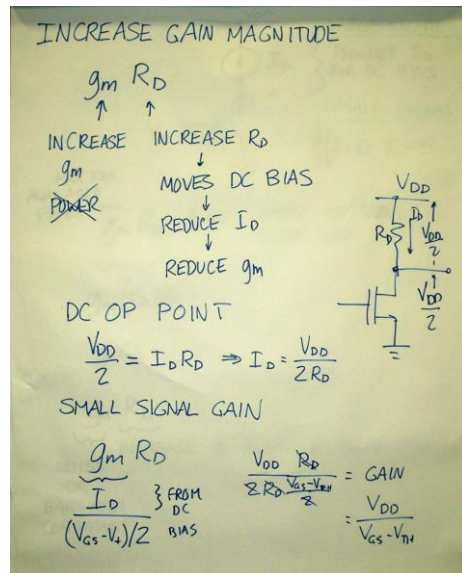
- Resistive load
- Limited to low gains



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Increasing Gain of Common Source Amplifier

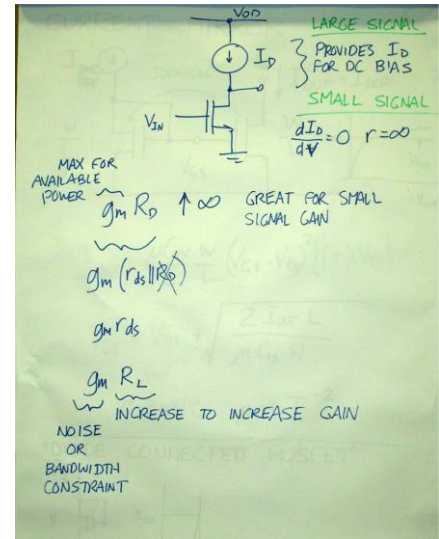
- Transconductance g_m is usually determined by power, bandwidth, or noise constraints.
- Problem: Increasing gain by increasing R_D affects DC bias point.
- Limited to low gains.



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Increasing Gain of Common Source Amplifier

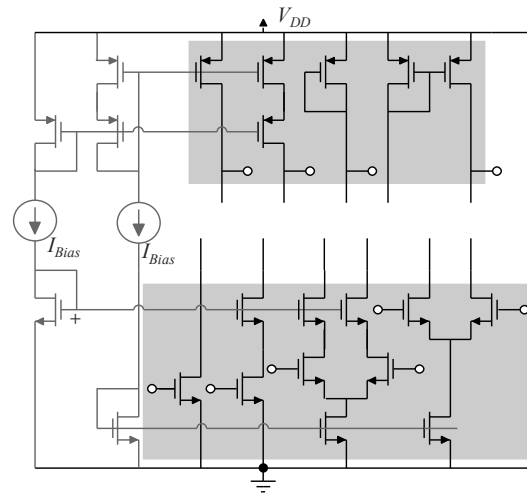
- Maximum R_D for increasing gain is open circuit $R_D = \infty$. But no DC bias current can flow.
- Is there a component that looks like an open circuit for small signals but allows DC current to flow?
- Current source.
- General principle: increase gain by increasing gain at output node of amplifier.



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Examples: Active-Load Amplifier

The combination of active load and transconductor forms an active-load amplifier. There are many different forms of both. Some examples of active-load amplifiers (both PMOS and NMOS configurations) are shown in the shaded areas.



Examples of active loads

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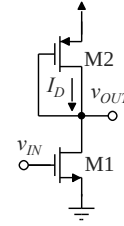
Active-Load Large-Signal Behavior

- Maximum output voltage, $v_{OUT(max)}$ (Ignore subthreshold current influence on the MOSFET):

$$v_{OUT(max)} \cong V_{DD} - |V_{TP}|$$

- Minimum output voltage, $v_{OUT(min)}$ (Assume that M_1 is nonsaturated and that $V_{T1} = |V_{T2}| = V_T$):

$$v_{DS1} \geq v_{gs1} - V_{TN} \rightarrow v_{OUT} \geq v_{IN} - 0.7V$$



Gain stage with active load

- The current through M_1 is

$$i_D = \beta_1 \left((v_{GS1} - V_T) v_{DS1} - \frac{v_{DS1}^2}{2} \right) = \beta_1 \left((v_{DD} - V_T) v_{OUT} - \frac{(v_{OUT})^2}{2} \right)$$

- And the current through M_2 is

$$i_D = \frac{\beta_2}{2} (v_{SG2} - V_T)^2 = \frac{\beta_2}{2} (V_{DD} - v_{OUT} - V_T)^2 = \frac{\beta_2}{2} (v_{OUT} + V_T - V_{DD})^2$$

- Equating these currents gives the minimum v_{OUT} as

$$v_{OUT(min)} \cong V_{DD} - V_T - \frac{V_{DD} - V_T}{\sqrt{1 + (\beta_2/\beta_1)}}$$

where and

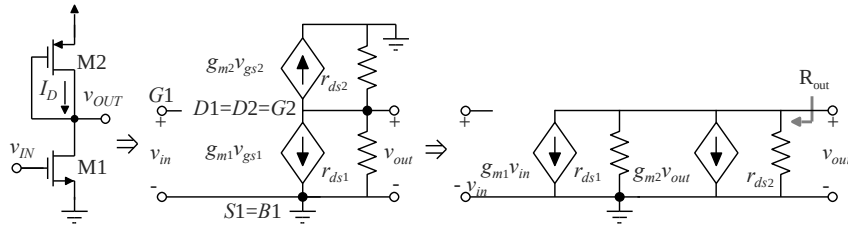
$$\beta_1 = \frac{W_1 \mu_0 C_{ox}}{L_1}$$

$$\beta_2 = \frac{W_2 \mu_0 C_{ox}}{L_2}$$

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Active-Load Small-Signal Behavior

- The development of the small-signal model for the active-load inverter is shown in the following:



Basic active-load amplifier with equivalent models

- Sum the currents at the output node to get

$$g_{m1}v_{in} + g_{ds1}v_{out} + g_{m2}v_{out} + g_{ds2}v_{out} = 0$$

- Solving for the voltage gain gives

$$\frac{v_{out}}{v_{in}} = \frac{-g_{m1}}{g_{ds1} + g_{ds2} + g_{m2}} \cong -\frac{g_{m1}}{g_{m2}} = -\left(\frac{k'_n W_1 L_2}{k'_p L_1 W_2}\right)^{1/2}$$

- The small-signal output resistance can also be found from the previous by letting $v_{in} = 0$ to get

$$R_{out} = \frac{1}{g_{ds1} + g_{ds2} + g_{m2}} \cong \frac{1}{g_{m2}}$$

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Active-Load Frequency Response

- The figure on the right shows the incorporation of the parasitic capacitors into the small-signal model.
- If you assume the input voltage has a small source resistance, then you can write the following:

$$sC_M(V_{out} - V_{in}) + g_m V_{in} + G_{out} V_{out} + sC_{out} V_{out} = 0$$

$$\therefore V_{out}(G_{out} + sC_M + sC_{out}) = -(g_m - sC_M)V_{in}$$

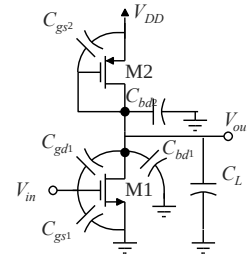
$$\frac{V_{out}}{V_{in}} = \frac{-(g_m - sC_M)}{G_{out} + sC_M + sC_{out}} = -g_m R_{out} \left[\frac{1 - \frac{sC_M}{g_m}}{1 + sR_{out}(C_M + C_{out})} \right] = \frac{-g_m R_{out} \left(1 - \frac{s}{z_1} \right)}{1 + \frac{s}{p_1}}$$

Where:

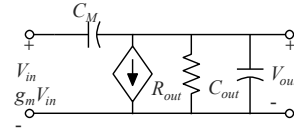
$$g_m = g_{m1}, \quad p_1 = \frac{-1}{R_{out}(C_{out} + C_M)} \quad \text{and} \quad z_1 = \frac{g_{m1}}{C_M}$$

and

$$R_{out} = [g_{ds1} + g_{ds2} + g_{m2}]^{-1} \cong \frac{1}{g_{m2}}, \quad C_M = C_{gd1}, \quad \text{and} \quad C_{out} = C_{bd1} + C_{bd2} + C_{gs2} + C_L$$



Schematic

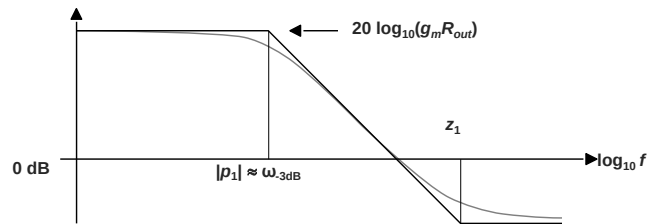


Model

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Active-Load Frequency Response (continued)

If $|p_1| < z_1$, then the -3 dB frequency is approximately equal to $[R_{out}(C_{out}+C_M)]^{-1}$



Idealized Bode plot

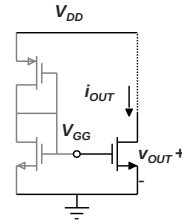
Note: The poles in a MOSFET circuit can be found by summing the capacitance connected to a node and multiplying this capacitance times the equivalent resistance from this node to ground and inverting the product.

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MOS Current Sinks and Sources

A sink or source is characterized by two quantities:

- r_{out} : A measure of the flatness of the current sink or source (its independence of voltage).
- V_{MIN} : The minimum voltage across the sink or source for which the current is no longer constant.

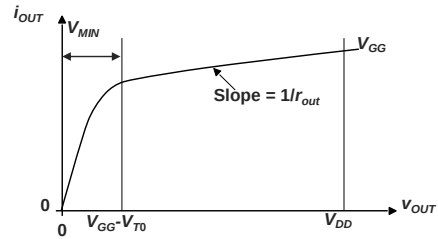


Simple current sink

$$r_{out} = \frac{1}{d_{iD} / dv_{DS}} = \frac{1 + \lambda V_{DS}}{\lambda I_D} \approx \frac{1}{\lambda I_D}$$

and

$$V_{MIN} = V_{DS}(\text{sat}) = V_{GS} - V_{TO} = V_{GG} - V_{TO}$$



Basic current sink I/V curve

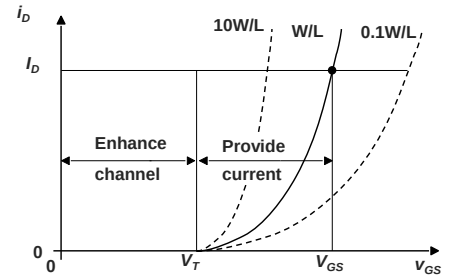
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Gate-Source Voltage Components

The gate-source voltage consists of two parts:

- Creating a channel
- Causing current flow

Notice that V_{MIN} can be reduced by using large values of W/L .



Voltage components of current sink current

$$V_{GS} = V_{TO} + V_{ON} = \text{part to enhance the channel} + \text{part to cause current flow}$$

Where:

$$V_{ON} = V_{DS}(\text{sat}) = V_{GS} - V_{TO}$$

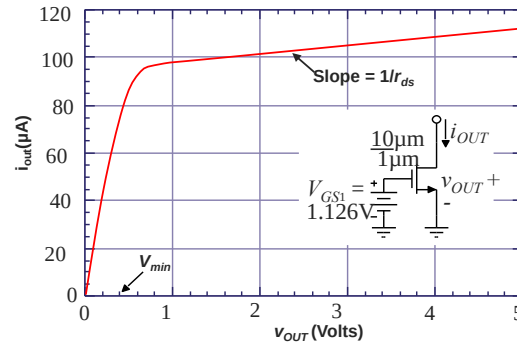
$$\therefore V_{MIN} = V_{ON} = V_{DS}(\text{sat}) = \sqrt{\frac{2I_D}{k'(W/L)}} \quad \text{for the simple current sink}$$

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Example: MOS Current Source or Sink

Several issues to consider with current sources or sinks:

- V_{MIN} : You want V_{MIN} to approach zero or at least approach $V_{ds(sat)}$.
- Slope: You want to keep R_{out} as high as possible (that is, slope should be as close to zero as possible) so that the current is as independent from the output voltage as possible.
- MOS transistors have a smooth transition at V_{min} compared to BJT. $1/I$ is called *early voltage* (here 25V – quite large, that is, would be smaller for very short transistors).



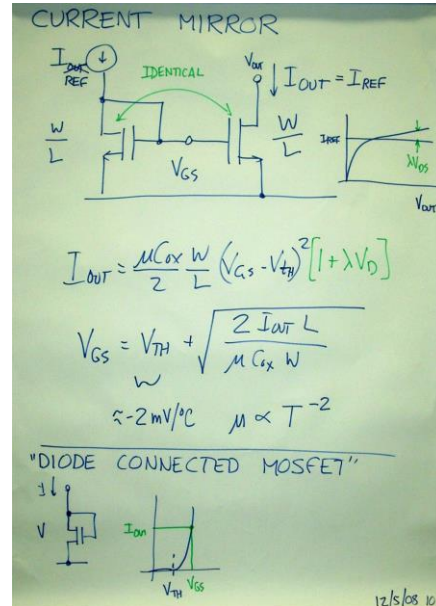
Current sink schematic and response

Example: $(k'_n = 110 \mu A/V^2, V_T = 0.7V \text{ and } \lambda = 0.04V^{-1})$
 $\Rightarrow r_{ds} = 250k\Omega$

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Current Mirror Principle

- Where did $V_{GS} = 1.126$ V come from?
- If fixed voltage: will be incorrect over PVT variations.
- General IC design principle:
 - Identical devices
 - Under identical conditions
 - Show identical behavior
- Force diode connected MOSFET to carry desired I_{REF} current.
- Correct V_{GS} for $I_{OUT} = I_{REF}$
- Can scale W ratio for different current ratio.



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Cascode MOS Current Source or Sink

The addition of M2' increases the output resistance and, therefore, reduces the slope of the I_d/V_{ds} curve.

Add negative feedback to increase the resistance.

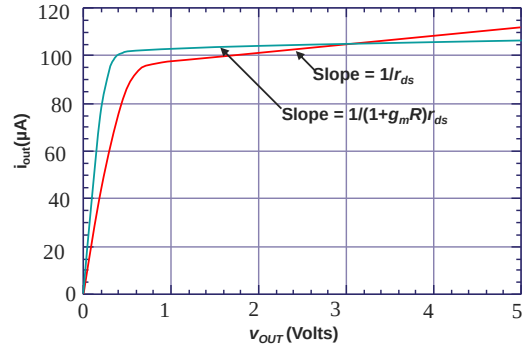
Large-signal intuitive view:

- Assume i_{OUT} increases.
- As a result, v_S increases.
- Because the gate is held constant at V_{GG} , then v_{GS} decreases.
- The decrease in v_{GS} causes i_{OUT} to decrease, opposing the original increase.

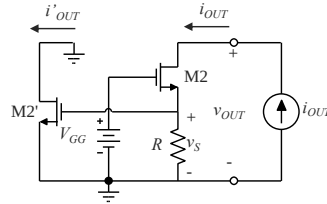
$$i_{OUT}' = g_m v_S = g_m R i_{OUT} \quad \therefore \text{Loop gain} = \frac{i_{OUT}'}{i_{OUT}} = g_m R$$

$$r_{out}(w.fb.) = r_{out}(w/o.fb.) \times [1 + g_m R] = r_{ds}(1 + g_m R)$$

$$\text{If } g_m R \gg 1 \text{ then } r_{out}(w.fb.) \approx g_m r_{ds} R$$



Cascode current sink response



Cascode current sink



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Cascode Output Resistance

Before looking at the cascode current source/sink, consider the effect of adding a source resistance from a small-signal perspective. (The previous slide dealt with the output resistance from a large-signal standpoint.)

Loop equation:

$$\begin{aligned} v_{out} &= (i_{out} - g_{m2}v_{gs2} - g_{mbs2}v_{bs2})r_{ds2} + i_{out}R \\ &= i_{out}(r_{ds2} + R) - g_{m2}r_{ds2}v_{gs2} - g_{mbs2}r_{ds2}v_{bs2} \end{aligned}$$

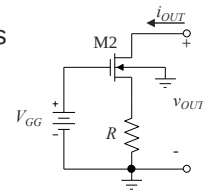
But:

$$v_{gs2} = 0 - v_{s2} = -i_{out}R \text{ and } v_{bs2} = 0 - v_{s2} = i_{out}R$$

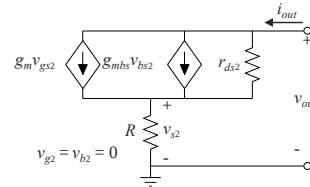
Therefore:

$$v_{out} = i_{out}[r_{ds2} + R + g_{m2}r_{ds2}R + g_{mbs2}r_{ds2}R]$$

$$r_{out} = \frac{v_{out}}{i_{out}} = r_{ds2} + R + g_{m2}r_{ds2}R + g_{mbs2}r_{ds2}R \approx g_{m2}r_{ds2}R = \mu_2 R$$



Small-signal equivalent circuit



Small-signal equivalent model

A general principle emerges:

The output resistance of a cascode circuit $\approx R \cdot$ (common source voltage gain of the cascoding transistor)

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Cascode Output Resistance (continued)

Now we can look at the output resistance when M1 is added in place of the source resistor:

Noting that $v_{gs1} = v_{g2} = v_{b2} = 0$ and writing a loop equation you get:

$$v_{out} = (i_{out} - g_{m2}v_{gs2}g_{mbs2}v_{bs2})r_{ds2} + r_{ds1}i_{out}$$

However:

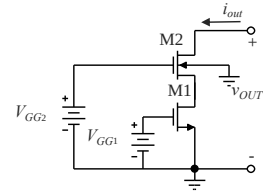
$$v_{gs2} = 0 - v_{s2} = -i_{out}r_{ds1} \text{ and } v_{bs2} = 0 - v_{s2} = -i_{out}r_{ds1}$$

Therefore:

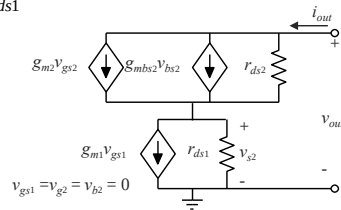
$$v_{out} = i_{out} [r_{ds1} + r_{ds2} + g_{m2}r_{ds1}r_{ds2} + g_{mbs2}r_{ds1}r_{ds2}]$$

$$r_{out} = \frac{v_{out}}{i_{out}} = r_{ds1} + r_{ds2} + g_{m2}r_{ds1}r_{ds2} + g_{mbs2}r_{ds1}r_{ds2}$$

$$\approx g_{m2}r_{ds1}r_{ds2} = \mu_2 r_{ds1}$$



Cascode current sink circuit



Cascode current sink equivalent model

Note: Compare this result to the current sink with a source resistor and notice that if $r_{ds1} = R$, then the result is the same.

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Practical Current Source Implementation

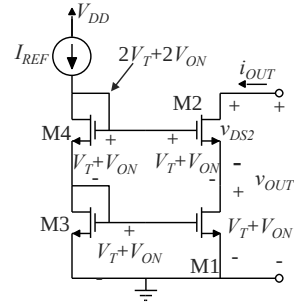
This implementation does not require batteries and uses the gate-source matching principle.

$$\begin{aligned} V_{MIN} &= V_T + V_{ON} + v_{DS2(\min)} = V_T + V_{ON} + V_{ON} \\ &= V_T + 2V_{ON} \end{aligned}$$

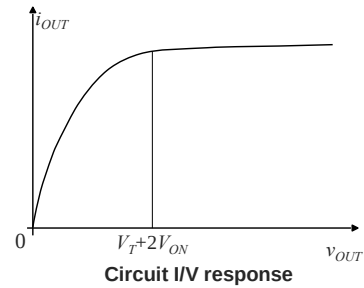
Assuming $I_{OUT} = 100 \mu\text{A}$ and $W2/L2 = W1/L1 = 10$ gives $V_{ON} = 0.426\text{V}$

$$\therefore V_{MIN} = 0.7\text{V} + 2(0.426\text{V}) = 1.55\text{V}$$

This is too high for low-supply applications.
This circuit should be used with low V_T transistors or supply voltages higher than 2.5V.



Practical cascode current source circuit



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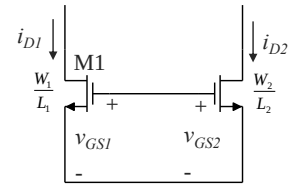
Gate-Source Matching Principle

If the gate-source voltages of two or more transistors are equal, and the transistors are matched and operating in the saturation region, then the currents are related by the W/L ratios of the individual transistors. The gate-source voltages can be directly connected or implied.

$$i_{D1} = \frac{k'W_1}{2L_1}(v_{GS1} - V_{T1})^2 \rightarrow (v_{GS1} - V_{T1})^2 = \frac{2k'i_{D1}}{(W_1/L_1)}$$

$$i_{D2} = \frac{k'W_2}{2L_2}(v_{GS2} - V_{T2})^2 \rightarrow (v_{GS2} - V_{T2})^2 = \frac{2k'i_{D2}}{(W_2/L_2)}$$

$$\text{If } v_{gs1} = v_{gs2}, \text{ then } \left(\frac{W_2}{L_2}\right)i_{D1} = \left(\frac{W_1}{L_1}\right)i_{D2} \text{ or } i_{D1} = \left(\frac{W_1/L_1}{W_2/L_2}\right)i_{D2}$$



Circuit for discussing the gate-source matching principle

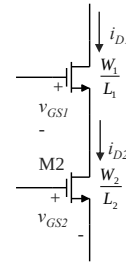
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Gate-Source Matching Principle (continued)

If the drain currents of two or more transistors are equal and the transistors are matched and operating in the saturation region, then the gate-source voltages are related by the W/L ratios (ignoring bulk effects).

$$\text{If } i_{D1} = i_{D2} \text{ , then } v_{GS1} = V_{T1} + \sqrt{\frac{W_2/L_2}{W_1/L_1}} (v_{GS2} - V_{T2})$$

$$\text{or if } W_2/L_2 = W_1/L_1 \text{ , then } v_{GS2} = v_{GS1}$$



Circuit for discussing
the gate-source matching principle

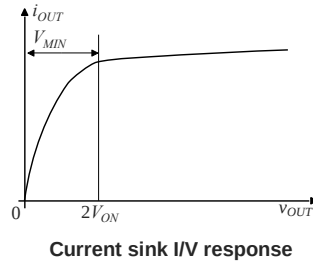
Note: V_{DS1} must equal V_{DS2} for ideal results.

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High-Swing Cascode Current Sink (continued)

Solution

- With $V_{MIN} = 1V$, choose $V_{ON} = 0.5V$.



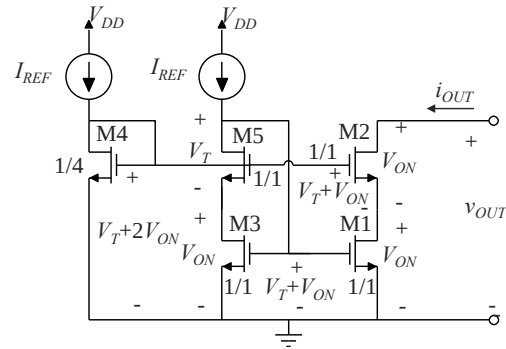
- Assuming M1 and M2 are identical gives.

$$\frac{W}{L} = \frac{2 \cdot I_{OUT}}{k' V_{ON}^2} = \frac{2 \cdot 100 \times 10^{-6}}{110 \times 10^{-6} \times 0.25} = 7.27 \Rightarrow \frac{W_1}{L_1} = \frac{W_2}{L_2} = \frac{W_3}{L_3} = \underline{\underline{7.27}} \text{ and } \frac{W_4}{L_4} = 1.82$$

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Improved High-Swing Cascode Current Sink

- Because the drain-source voltages of the matching transistors M1 and M3 are not equal in the previous design, $i_{OUT} \neq I_{REF}$.
- The improved design seeks to fix this problem.
- Notice that the drain-source voltage of M1 and M3 are identical, causing i_{OUT} to be a replication of I_{REF} .



Improved high-swing cascode circuit

Because $V_{MIN} = 2V_{ON} = V_{DS}(sat)$ let $V_{ON} = 0.5V_{MIN}$

$$V_{ON} = \sqrt{\frac{2I_{REF}}{k'(W/L)}} \Rightarrow \frac{W_1}{L_2} = \frac{W_2}{L_2} = \frac{W_3}{L_3} = \frac{W_5}{L_5} = \frac{2I_{REF}}{k'V_{ON}^2} = \frac{8I_{REF}}{k'V_{MIN}^2}$$

$$\frac{W_4}{L_4} = \frac{2I_{REF}}{k'(V_{GS4} - V_T)^2} = \frac{2I_{REF}}{k'(2V_{ON})^2} = \frac{I_{REF}}{2k'V_{ON}^2}$$

This is a good standard circuit.

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Self-Biased Cascode Current Sink

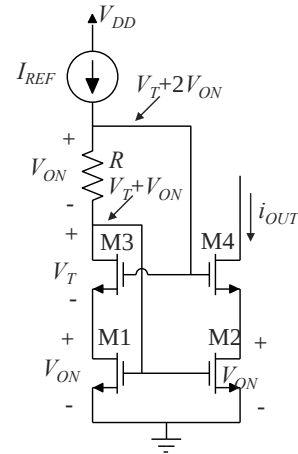
- The $V_T + 2V_{ON}$ bias voltage is developed through a series resistor.
- Design procedure:

Same as the previous high-swing cascode current sink except:

$$R = \frac{V_{ON}}{I_{REF}} = \frac{V_{MIN}}{2I_{REF}}$$

For the previous example:

$$R = \frac{0.3V}{2 \cdot 100\mu A} = 1.5k\Omega$$



Note: The circuit is acceptable, but R needs careful adjustment and can be large for low currents.

T.L. Brooks and A.L. Westwick, "A Low-Power Differential CMOS Bandgap Reference," *Proceedings of IEEE Integrated Solid-State Circuits Conference*, February 1994, pp. 248-249.

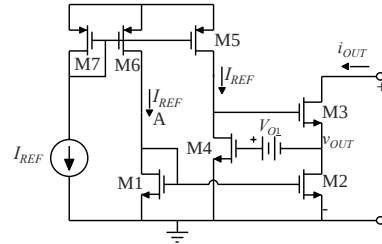
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Regulated Cascode Current Sink

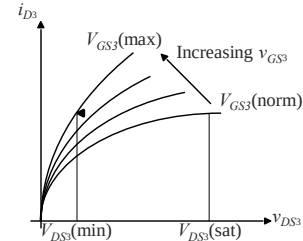
- Achieves very high output resistance by increasing the loop gain due to the M4-M5 inverting amplifier.

$$\begin{aligned}\text{Loop gain} &= g_{m3}r_{ds2} \left(\frac{g_{m4}}{g_{ds4} + g_{ds5}} \right) \\ &\approx \frac{g_{m3}r_{ds2}g_{m4}r_{ds4}}{2} \text{ if } r_{ds4} \approx r_{ds5} \\ \therefore r_{out} &\approx \frac{r_{ds3}g_{m3}r_{ds2}g_{m4}r_{ds4}}{2}\end{aligned}$$

- M3 maintains constant current even though it is no longer in the saturation region.
- Assume an i_{OUT} increase $\rightarrow v_{S3}$ increase $\rightarrow V_{GS4}$ increase $\rightarrow V_{G3}$ decrease $\rightarrow$ Large decrease in v_{GS3} $\rightarrow$ Large decrease in i_{OUT} .



Regulated cascode current sink circuit



Regulated cascode I/V response

E. Sackinger and W. Guggenbuhl, "A Versatile Building Block: The CMOS Differential Difference Amplifier," *IEEE J. of Solid-State Circuits*, vol. SC-22, no. 2, April 1987, pp. 287-294.

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Regulated Cascode Current Sink (continued)

Solving for the output resistance:

$$i_{out} = g_{m3}v_{gs3} + g_{ds3}(v_{out} - v_{gs4})$$

But:

$$v_{gs4} = i_{out}r_{ds2}$$

And:

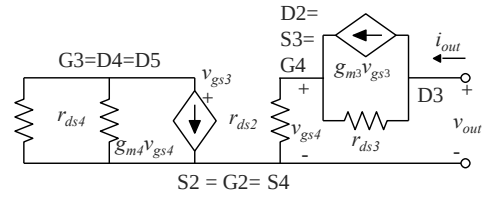
$$v_{gs3} = v_{gs} - v_{s3} = -g_{m4}(r_{ds4} \parallel r_{ds5})v_{gs4} - v_{gs4} = -r_{ds2}[1 + g_{m4}(r_{ds4} \parallel r_{ds5})]i_{out}$$

$$\therefore i_{out} = -g_{m3}r_{ds2}[1 + g_{m4}(r_{ds4} \parallel r_{ds5})]i_{out} + g_{ds3}v_{out} - g_{ds3}r_{ds2}i_{out}$$

$$v_{out} = r_{ds3}[1 + g_{m3}r_{ds2} + g_{m3}r_{ds2}g_{m4}(r_{ds4} \parallel r_{ds5})]i_{out}$$

$$\therefore r_{out} = \frac{v_{out}}{i_{out}} = r_{ds3}[1 + g_{m3}r_{ds2} + g_{ds3}r_{ds2} + g_{m3}r_{ds2}g_{m4}(r_{ds4} \parallel r_{ds5})]r_{ds3}g_{m3}r_{ds2}g_{m4}(r_{ds4} \parallel r_{ds5})$$

If $I_{REF} = 100 \mu A$, all W/Ls are $10 \mu m/1 \mu m$, you get $r_{ds} = 0.25 M\Omega$ and $g_m = 469 \mu S$ which gives $r_{out} \approx (0.25 M\Omega)(469 \mu S)(0.25 M\Omega)(469 \mu S)(0.125 M\Omega) = 1.72 G\Omega$



Regulated cascode current sink small signal model



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Regulated Cascode Current Sink (continued)

V_{MIN}

- Without the use of the V_{O1} battery shown, V_{MIN} is too high.
- $V_{MIN} = V_{GS4} + V_{DS3(sat)} = V_T + 2V_{ON}$

Minimizing V_{MIN}

- If $V_{O1} = V_T$, then $V_{MIN} = 2V_{ON}$
- If $V_{GS4A} - V_{GS4B} = V_{DS2(sat)} = V_{ON}$, then $V_{MIN} = 2V_{ON}$

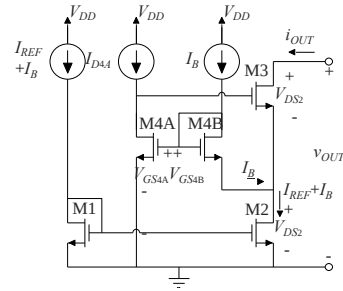
$$\therefore \sqrt{\frac{2I_{D4}}{K_N'(W_{4A}/L_{4A})}} - \sqrt{\frac{2I_B}{K_N'(W_{4B}/L_{4B})}} = \sqrt{\frac{2I_B + 2I_{REF}}{K_N'(W_2/L_2)}}$$

or

$$\sqrt{\frac{I_{D4}}{W_{4A}/L_{4A}}} - \sqrt{\frac{I_B}{W_{4B}/L_{4B}}} = \sqrt{\frac{I_B + I_{REF}}{W_2/L_2}}$$

Several solutions exist. For example, let $I_B = I_{REF}$ and gives $I_{D4A} = 5.824I_{REF}$ assuming all W/L ratios are identical.

Note: The circuit is already complex and is preferred only if output resistance really counts, for example, to maximize gain.

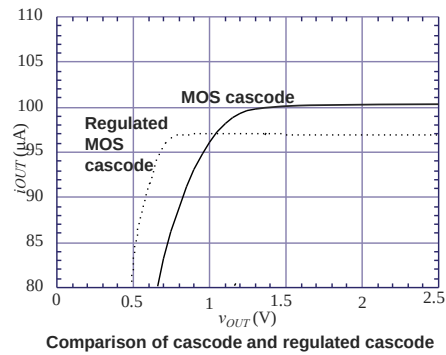


Regulated cascode current sink schematic

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Cascode vs. Regulated Cascode Current Sinks

- The regulated cascode current is smaller than the cascode current because the drain-source voltages of M1 and M2 are not equal.
- The regulated cascode current sink has a smaller V_{MIN} due to the fact that M3 can have a drain-source voltage smaller than $V_{DS(sat)}$.



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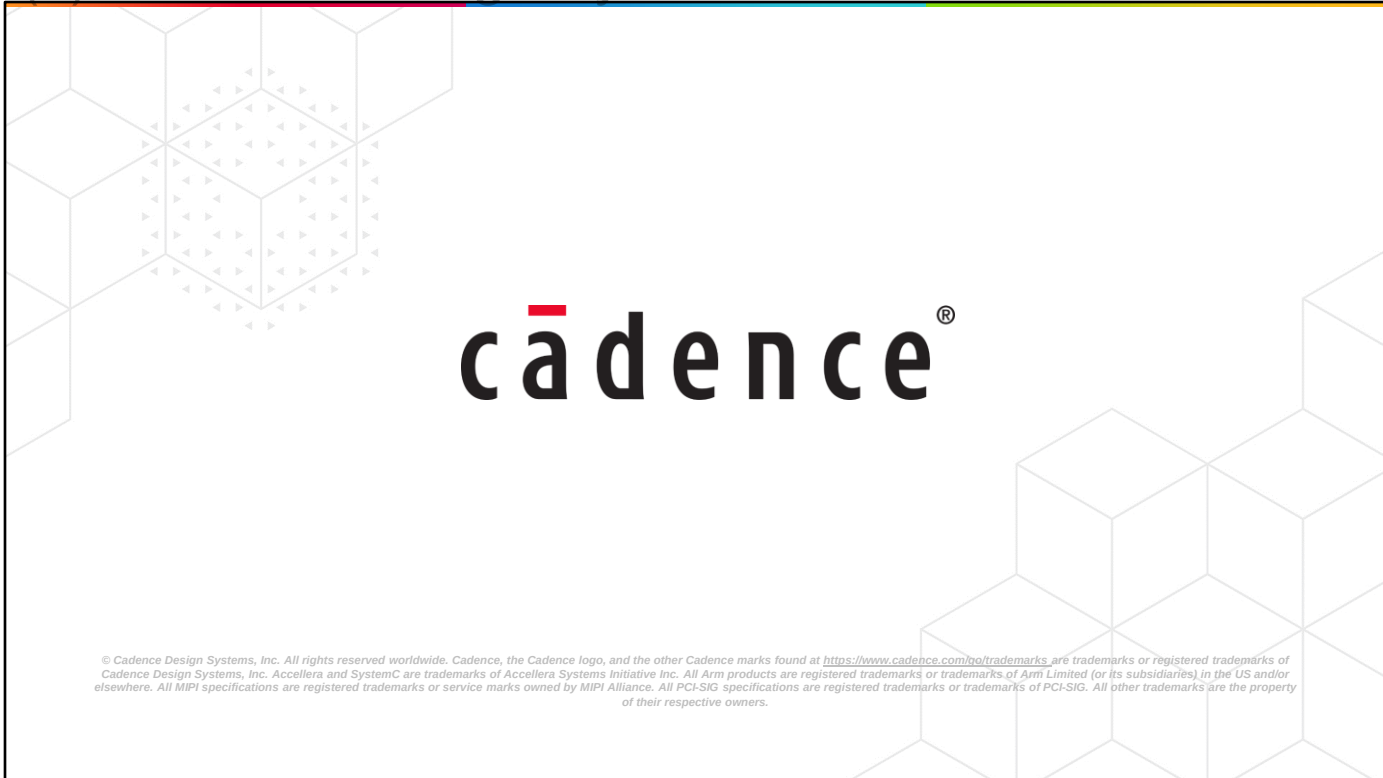
Summary: Active Loads

Current sink or source	r_{OUT}	V_{MIN}
Simple MOS current sink	$r_{ds} = \frac{1}{\lambda I_D}$	$V_{DS(sat)} = V_{ON}$
Cascode MOS	$\approx g_{m2} r_{ds2} r_{ds1}$	$V_T + 2V_{ON}$
Minimum V_{MIN} cascode current sink	$\approx g_{m2} r_{ds2} r_{ds1}$	$2V_{ON}$
Regulated cascode current sink*	$\approx r_{ds3} g_{m3} r_{ds2} g_{m4} (r_{ds4} \parallel r_{ds5})$	$\approx V_T + V_{ON}$
Minimum V_{MIN} regulated cascode current sink*	$\approx r_{ds3} g_{m3} r_{ds2} g_{m4} (r_{ds4} \parallel r_{ds5})$	$\approx V_{ON}$

*Unfortunately, the regulated cascode current sink has a dominant pole in the feedback loop, which can cause a pole-zero doublet that leads to a combination of fast and slow time constants. For this reason, the regulated cascode circuit should be used only in biasing applications where the impact of this dynamic is understood.



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